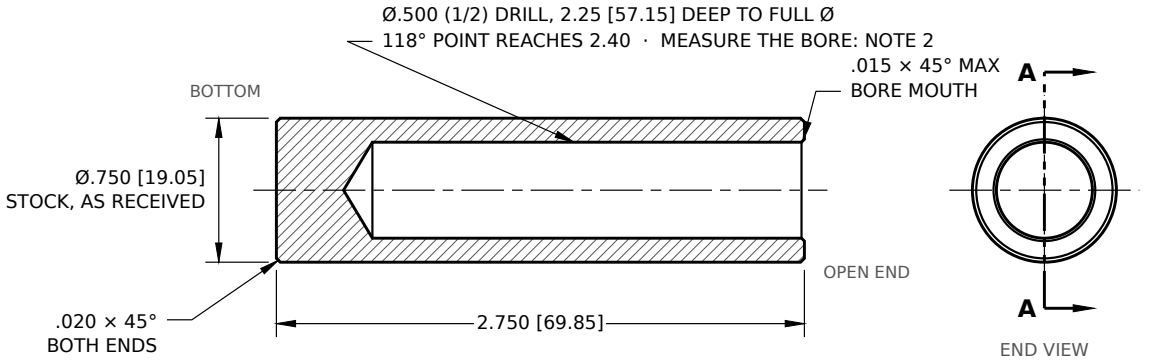
Charge cup and plug, as made

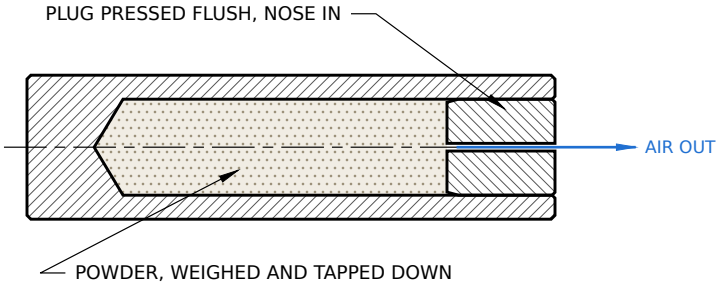
The sizes of the cups and plugs made for the first runs (#222), so new ones match them. Every part is drawn cut through the middle; hatching is metal.

REV	DESCRIPTION	DATE
A	FIRST ISSUE: THE AS-MADE SIZES (#222, PR #277)	2026-10-19



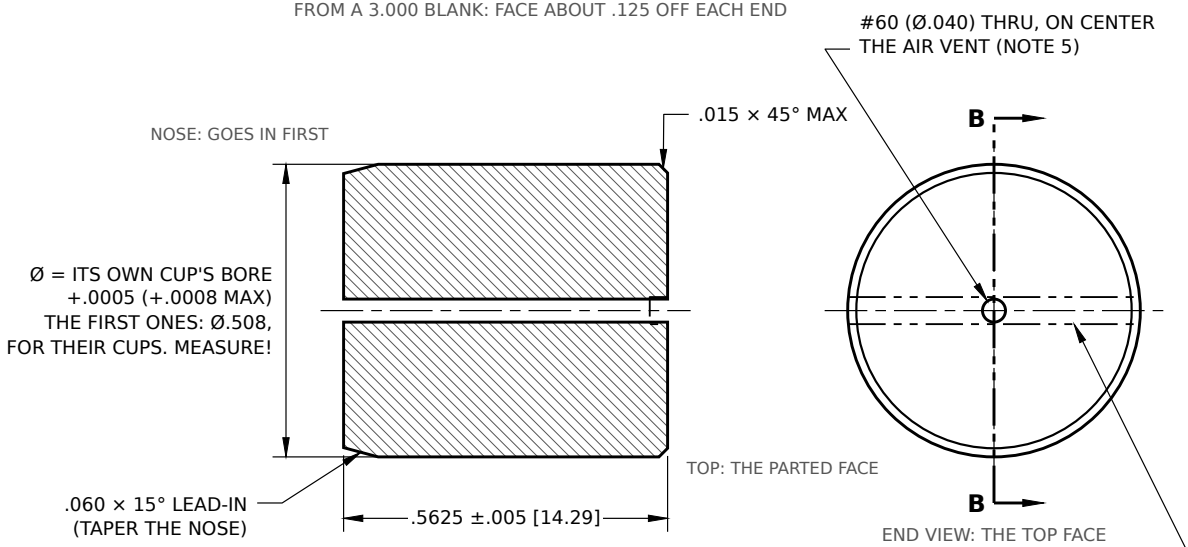
CUP (P2) · SECTION A-A · 1:1

FROM A 3.000 BLANK: FACE ABOUT .125 OFF EACH END



ASSEMBLED AND FILLED · REF · 1:1

STANDS PLUG-UP IN THE CRUCIBLE



PLUG (P3) · SECTION B-B · 3:1

ONE PER CUP, MADE AFTER ITS CUP

NOTES

1. MATERIAL: 6063-T52 ALUMINUM ROD, Ø3/4 (McMASTER 1640T16), FROM THE 3.000 SAWN BLANKS. LEAVE THE OD AS RECEIVED.
2. MAKE THE CUP FIRST. MEASURE ITS BORE IN THE TOP 1/2 WITH A TELESCOPING GAGE AND A MICROMETER, TWICE, 90° APART, AND TURN ITS PLUG TO THE AVERAGE +.0005. NEVER OVER +.0008: A TIGHTER PLUG PERMANENTLY STRETCHES THE CUP WALL.
3. ONE PLUG PER CUP: THEY ARE NOT INTERCHANGEABLE, AND A PLUG GOES INTO ITS CUP ONCE, AFTER FILLING. BAG EACH PAIR TOGETHER AND LABEL THE BAG. NO MARKER, SCRIBING OR STAMPS ON THE PARTS.
4. ORDER OF OPERATIONS, TOOLS, SPEEDS AND THE RECORD SHEET: SEE THE PROCESS SHEETS, atomizer-charge/process-sheet/ (PR #277).
5. EVERY PLUG GETS THE #60 VENT. THE FURNACE IS PUMPED DOWN BEFORE MELTING, AND WITHOUT A VENT THE AIR UNDER THE PLUG HAS TO ESCAPE THROUGH THE POWDER.
6. CUT DRY OR WITH LIGHT OIL. NO WATER-SOLUBLE COOLANT, AND NO FILES, EMERY OR SCOTCH-BRITE: RESIDUE AND GRIT END UP IN THE MELT. BRUSH STEEL AND BRASS CHIPS OFF THE LATHE FIRST.
7. FLUSH THE BORE WITH 99% IPA, SWAB IT AND FLUSH AGAIN; DRY AT 80 °C FOR 30 MIN. WEIGH EACH PART TO 0.01 g AND WRITE THE MASSES ON THE BAG.
8. HYDRAULIC-PRESS RUNS ONLY: DRILL 31/64, THEN REAM TO .500 SO THE BORE IS STRAIGHT AND SMOOTH, AND CUT THE CHANNEL. THAT PLUG SLIDES IN RATHER THAN PRESSING IN, AND ITS CLEARANCE ISN'T SET YET: ASK STERLING BEFORE MAKING ONE.

TWO VERSIONS: ALL ELSE IS THE SAME

	STANDARD, AS MADE (SHOWN)	THIN WALL
FOR	ALL RUNS SO FAR	HIGH-POWDER RUNS (#264)
CUP BORE	Ø.500 (1/2) DRILL	Ø.625 (5/8) DRILL, AFTER THE 1/2
CUP WALL	.125	.0625
BORE DEPTH	2.25 TO FULL Ø (POINT AT 2.40)	2.25 TO FULL Ø (POINT AT 2.44)
PLUG LENGTH	.5625 (9/16)	.250 (1/4)
PLUG Ø	CUP BORE +.0005 (+.0008 MAX)	CUP BORE +.0005, NO MORE
MASS, CUP / PLUG (REF)	≈34 g / 5.0 g	≈22 g / 3.4 g

MASSES: 6063 AT 2.69 g/cm³, NOMINAL STOCK. THE BAR'S OD TOLERANCE MOVES THEM ABOUT 4 %.

BYU VERTICAL CLOUD LAB · rePOWDER ATOMIZER CHARGE

CHARGE CUP AND PLUG, Ø3/4 6063

DWG NO.	REV	SHEET	PROJECTION
AC-CUP-PLUG	A	1 OF 1	

SCALE	UNITS
CUP 1:1, PLUG 3:1, ON LETTER AT 100 %	INCH [mm]

MATERIAL · FINISH
6063-T52 ALUMINUM ROD, Ø3/4 (McMASTER 1640T16) · AS MACHINED, DEGREASED

UNLESS OTHERWISE SPECIFIED
.XX ±.01 .XXX ±.005 ANGLES ±1° BREAK SHARP EDGES .005-.015 DO NOT SCALE

DRAWN	CHECKED
CLAUDE (AI) FOR R. GUYMON, 2026-10-19	

HOW TO MAKE THEM: atomizer-charge/process-sheet/ · THIS SHEET'S SOURCE: atomizer-charge/drawing/cup_plug.py